

Title: Developing daily gap-filled ocean color datasets using deep neural networks: Going from proof-of-concept to production

PI: Elizabeth Eli Holmes, PhD

Short summary:

Ocean color (OC) remote sensing is critical for understanding ocean productivity and marine ecosystems, and the new hyperspectral satellite missions greatly expand its applications.

However, OC sensors cannot penetrate clouds, leaving large data gaps across the oceans. We developed and validated a deep-learning gap-filling approach using the data themselves to create training data. We need help to scale the method from a regional prototype to a global product and an operational workflow.

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Introduction

Satellite ocean color data are widely used to estimate phytoplankton biomass and ocean productivity, which are foundational variables for understanding marine ecosystems, fisheries dynamics, and the global carbon cycle. In Feb 2024, NASA's Phytoplankton, Aerosols, Clouds and ocean Ecosystems (PACE) mission came on-line and this provides new hyperspectral ocean color data; we went from 7-12 spectra to 172 spectra giving us the full color spectra. This has opened many new applications and products, for example estimates of phytoplankton size structure from space. However, satellite sensors cannot penetrate clouds, resulting in missing observations over large areas of the ocean. Because clouds cover much of the Earth at any given time, these gaps can significantly limit the usefulness of ocean color data for scientific and operational applications.

Traditional gap-filling methods for ocean color rely primarily on spatial interpolation from nearby observations. These approaches can perform poorly when large contiguous regions are missing and often smooth spatial features. Ocean color fields reflect dynamic biological and physical processes and commonly exhibit coherent spatial structures such as fronts, filaments, and eddies. Preserving these spatial patterns is critical for scientific interpretation and downstream modeling.

Over the last 3 years, our research group has developed a proof-of-concept deep-learning approach for gap-filling ocean color data using a convolutional neural network (U-Net architecture) similar to Zhang & Zhou (2023) and Hong et al (2023). The model integrates co-located environmental variables from satellite and reanalysis products, including sea surface temperature, winds, and surface currents, along with temporal context from adjacent days. Training data are generated from the gappy original data themselves using a masking strategy that simulates realistic cloud cover patterns (Figure 1), allowing the model to reconstruct missing regions from the partially observed scenes (Sinha, Hang, Holmes 2024).

We have successfully implemented this approach for one variable: chlorophyll-a, a derived variable from multi-spectral remote-sensing. For this work, we focused on the North Indian Ocean, a region characterized by strong seasonal upwelling and frequent cloud cover during peak phytoplankton bloom periods. For this work, we developed a machine-learning-ready dataset for the North Indian Ocean containing harmonized variables and training masks. Cross-validation results of our UNet approach for gap-filling chlorophyll-a demonstrate that model predictions perform well (predictions match the non-missing remote-sensing values) and the spatial features are captured.

Why the eScience Accelerator program?

The next phase of this work requires transitioning from a research prototype to a scalable and reusable workflow capable of producing global gap-filled datasets. While our team has experience developing and validating deep-learning models and building cloud-optimized datasets, our work to date has been limited to single-GPU, in-memory training and relatively small spatial domains. Scaling to global datasets introduces new challenges in designing

efficient training pipelines for larger-than-memory data, implementing distributed or multi-GPU workflows, and developing robust cloud-based architectures for model training and inference. In addition, we hope to get guidance on approaches for tuning global models to regional conditions.

Project Objectives

Objective 1 — Scale our model-training workflows from regional to global and cloud-hosted datasets Develop and test training and inference pipelines capable of processing cloud-hosted global ocean color datasets that exceed memory limits. This includes implementing efficient batching strategies and evaluating distributed or multi-GPU approaches for model training.

Objective 2 — Develop approaches for regional optimization and model generalization Evaluate methods for adapting models trained on global data to regional ocean conditions, including proof-of-concept approaches for regional fine-tuning. This objective addresses the practical need for regionally optimized gap-filling across diverse ocean environments.

Product 1: Objectives 1 & 2 will be used to produce a gap-filled chlorophyll-a Zarr store (or similar) as our first product for the PACE community.

Objective 3 — Develop a proof-of-concept approach for gap-filling hyperspectral (multi-variable) ocean color Gap-filling the full 172-spectra remote-sensing reflectance data is the long-term goal. We don't expect to necessarily implement this during summer 2026, but hope to outline a workable strategy.

Broader Impacts

This project directly supports the development of applications for NASA's new hyperspectral ocean color mission (PACE), particularly in fisheries and ocean ecosystem research. The PI is actively involved in supporting these applications and serves as one of the organizers of the Fisheries and Ocean Ecosystems PACE ("Fish-PACE") User Group under the NASA PACE Applications program. Within this community, the extensive gaps in hyperspectral ocean color data caused by cloud cover are a major barrier to developing new scientific and operational applications. By promoting the development of gap-filling approaches for PACE data, this project will help enable broader use of hyperspectral datasets.

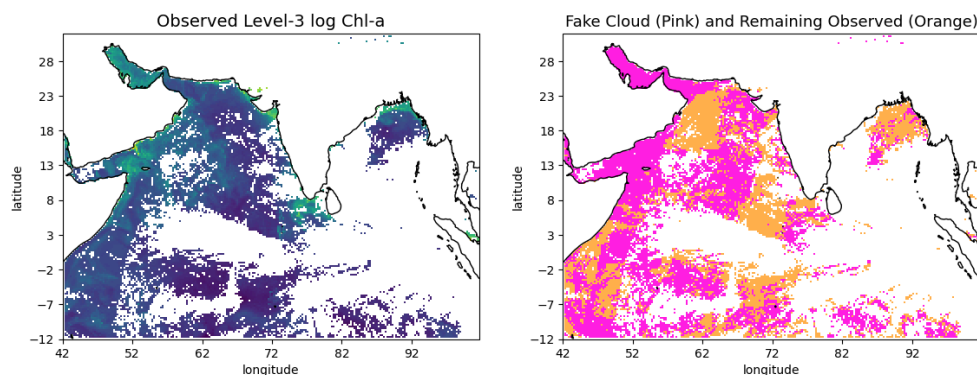


Figure 1: Application of masking using clouds 10 days in the future. This creates masks that are realistic in size and location. Work by Yifei Hang, UW and EE Holmes, NOAA.

References

Hong, Z., Long, D., Li, X., Wang, Y., Zhang, J., Hamouda, M. A., and Mohamed, M. M. 2023. A global daily gap-filled chlorophyll-a dataset in open oceans during 2001–2021 from multisource information using convolutional neural networks, *Earth Syst. Sci. Data*, 15, 5281–5300, <https://doi.org/10.5194/essd-15-5281-2023>.

Sinha, Shridhar; Hang, Yifei; Holmes, Elizabeth (2024). Developing daily gap-filled chlorophyll-a datasets using deep neural networks with co-located environmental variables. figshare. Presentation. <https://doi.org/10.6084/m9.figshare.27936036.v1>

Zhang X, Zhou M. A General Convolutional Neural Network to Reconstruct Remotely Sensed Chlorophyll-a Concentration. 2023. *Journal of Marine Science and Engineering*. 11(4):810. <https://doi.org/10.3390/jmse11040810>

Questions

Describe what is measured, the formats or file types currently used to store the data, and the approximate size of the data (e.g. 100 MB, 20 GB, 3 TB).*

The primary data are satellite-derived ocean color measurements, specifically remote-sensing reflectance (Rrs) across multiple spectral bands. For the current proof of concept, we are using derived single variable ocean color products, including chlorophyll-a (Chl-a) and Apparent Visible Wavelength (AVW). The data are stored as NetCDF files and are publicly available in cloud-hosted repositories maintained by NOAA and NASA. I have extensive experience with the data.

Our current cloud-native test dataset (Zarr) for the North Indian Ocean is approximately 100 GB and publicly cloud hosted. The datasets required to scale this work to global coverage are expected to be multiple terabytes (TB) in size, depending on spatial resolution, number of spectral bands, and temporal coverage. Hopefully by the start of the project we will have VirtualiZarr versions of our global dataset ready.

Which parts of the project require data science and/or AI expertise beyond your team's capacity or knowledge base?

The primary areas requiring data science and AI expertise involve scaling deep-learning workflows from a regional proof of concept to global production. We have experience developing and validating gap-filled ocean color data using CNN models, and we have experience creating gridded cloud-optimized 3D (time/lat/lon) datasets. But the work to date has been limited to single-GPU, in-memory training and relatively small spatial domains (Indian Ocean).

We need guidance on designing efficient training pipelines for larger-than-memory datasets, including batching strategies, distributed or multi-GPU training, and cloud-based model deployment. We also want to discuss workflow architecture for these larger global-scale data processing and model inference. Lastly, we would benefit from advice on handling multi-spectral hyperspectral data and for adapting global models to regional conditions.

These areas represent the key steps required to move from a successful research prototype to an operational workflow.

What tools have you used and hope to use?

So far, we have used Python-based deep learning to train 2D U-Net and CNN models using TensorFlow/Keras. We used xarray and Zarr to create a cloud-native, machine-learning-ready

dataset for the North Indian Ocean for our proof of concept work. Model development so far has been limited to single-GPU, in-memory training workflows.

During the Accelerator, we hope to develop methods for larger-than-memory training and scalable inference on global ocean color datasets. I think this will include tools such as xbatcher or similar approaches for generating training batches from xarray/Zarr data, distributed or multi-GPU training strategies, and cloud architecture for managing large model training and prediction workflows.

How would this Accelerator project fit into your broader research portfolio?*

I am heavily involved in supporting fisheries and ocean ecosystem applications for the new hyperspectral ocean color mission (PACE). I am one of the organizers for the new Fisheries and Ocean Ecosystems PACE ("Fish-PACE") User Group under the NASA PACE Applications program. The extensive gaps (clouds) in the hyperspectral data are a major barrier for developing applications. Researchers need a practical way to fill gaps AND to make it easier for researchers to do regionally-optimized gap-filling.

If successful, this project will produce publicly available datasets, software, and reproducible workflows that support operational gap-filling of satellite ocean color data, with a focus on PACE ocean color gap-filling. All resulting products, publications, and presentations will properly acknowledge the role of the eScience Institute Data Science and AI Accelerator.

Product 1 — Global gap-filled chlorophyll-a dataset

By the end of the project period, we will produce a daily global gap-filled chlorophyll-a (Chl-a) dataset derived from the PACE Level-3 product (February 2024 to present). The dataset will be stored in cloud-native Zarr format and publicly served for the PACE community, with the goal of distribution through the NASA PACE Applications Program as an experimental product.

Product 2 — Proof of concept for hyperspectral gap-filling

We will develop a proof of concept workflow for gap-filling multi-band remote-sensing reflectance (approximately 172 spectral bands). This represents the long-term goal of reconstructing complete hyperspectral ocean color datasets in the presence of cloud cover. This would represent a truly novel and unavailable product for ocean color applications.

Product 3 — Prototype Python package

We will develop a proof-of-concept open-source Python package that allows users to apply the trained model to gap-fill ocean color data in the cloud. Ultimately allowing users to gap-fill their specific time/region is more efficient than producing new daily global gap-filled Zarr stores.

Publications and presentations

We expect to produce a peer-reviewed publication and present results at the AGU Fall Meeting (December 2026), with follow-on publications as the hyperspectral workflow matures.